Modeling Longitudinal Change in Brain Signals of Twins

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Outline of Talk

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- Overview of Model
- Preliminary Results

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- Future Directions

Overview of Data

Data from Minnesota Twin Family Study*

Data from approximately 750 twin pairs.

- About 480 MZ pairs, and 270 DZ pairs
- Similar number of males and females

Assessed at target ages of 11 (intake), 14, 17, 20, and 24.

65% with 4 or 5 measurements, 83% with at least 3 measurements.

Record ERP data from same task at each measurement occasion.

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*https://mctfr.psych.umn.edu/

ERPs from Begleiter Rotated Heads Task

240 total trials

- 160 are non-target (subjects ignore)
- 80 are targets (subject indicates left or right ear)

Data averaged over 60-80 target trials for each subject

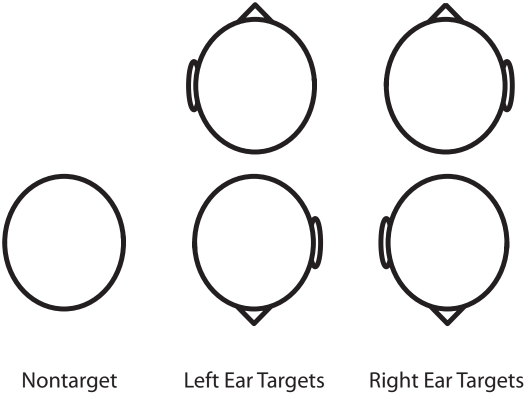

Figure 1: Stimuli from rotated heads task.

Electrode Montage

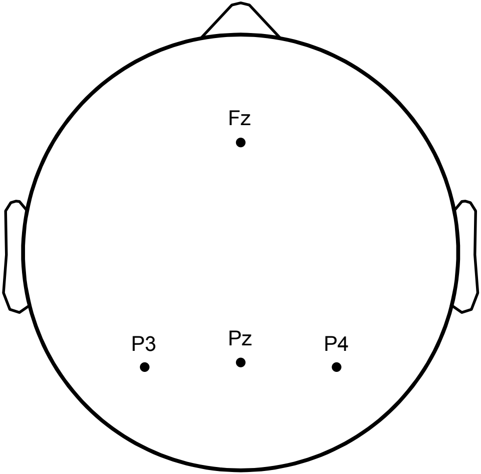

Figure 2: Electrode montage for MTFS data.

Classic ACE Model

Classic ACE Model for Twin Data

Let $y_{ij} \in \mathbb{R}$ denote the response variable recorded from the j-th twin of the i-th twin pair for $j \in \{1, 2\}$ and $i \in \{1, \ldots, n\}$.

The standard ACE model for twin data has the form

$$y_{ij} = \mu + a_{ij} + c_i + e_{ij} \tag{1}$$

where

- $\mu \in \mathbb{R}$ is the unknown mean
- $a_{ij} \sim \mathrm{N}(0, \sigma_A^2)$ are the latent *additive genetic* effects
- $c_i \overset{\text{iid}}{\sim} \mathrm{N}(0, \sigma_C^2)$ are the latent *common environment* effects
- $e_{ij} \overset{\text{iid}}{\sim} \mathrm{N}(0, \sigma_E^2)$ are the latent *unique environment* effects
- a_{ij}, c_i, and e_{ij} are mutually independent of one another

Covariance Structure of Additive Genetic Effects

The assumed covariance structure for the additive genetic component is

$$\mathrm{Cov}(a_{i1}, a_{i2}) = \begin{cases} \sigma_A^2 & \text{if } z_i = 1 \text{ (monozygotic)} \\ \frac{1}{2}\sigma_A^2 & \text{if } z_i = 2 \text{ (dizygotic)} \end{cases}$$

where $z_i \in \{1, 2\}$ denotes the number of zygotes for the i-th twin pair.

Parameterize the effects as $\mathbf{a}_i = \mathbf{\Phi}_{z_i} \tilde{\mathbf{a}}_i$ where $\tilde{\mathbf{a}}_i \overset{\text{iid}}{\sim} \mathrm{N}(\mathbf{0}, \sigma_A^2 \mathbf{I})$ and

$$\mathbf{\Phi}_1 = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \qquad \text{and} \qquad \mathbf{\Phi}_2 = \begin{pmatrix} 1 & 0 \\ 1/2 & \sqrt{3/4} \end{pmatrix}$$

are the corresponding matrices for MZ and DZ twins.

Estimating the Mean and Variance Parameters

Letting $\mathbf{y}_i = (y_{i1}, y_{i2})'$, the model for the i-th twin pair can be written as

$$\mathbf{y}_i = \mu\mathbf{1} + \mathbf{Z}_i\boldsymbol{\gamma}_i + \mathbf{e}_i$$

where $\mathbf{Z}_i = [\boldsymbol{\Phi}_{z_i}, \mathbf{1}]$ and $\boldsymbol{\gamma}_i = (\tilde{a}_{i1}, \tilde{a}_{i2}, c_i)'$.

Given $\{\sigma_A^2, \sigma_C^2, \sigma_E^2\}$, it is simple to estimate $\hat{\mu}$, $\hat{\boldsymbol{\gamma}}_i$, and $\hat{\mathbf{e}}_i$.

Can estimate the variance components using ML or REML estimation with the constraint that $\mathrm{Var}(\tilde{a}_{i1}) = \mathrm{Var}(\tilde{a}_{i2})$.

ACE Estimated Mean Structure: $\hat{\mu}$

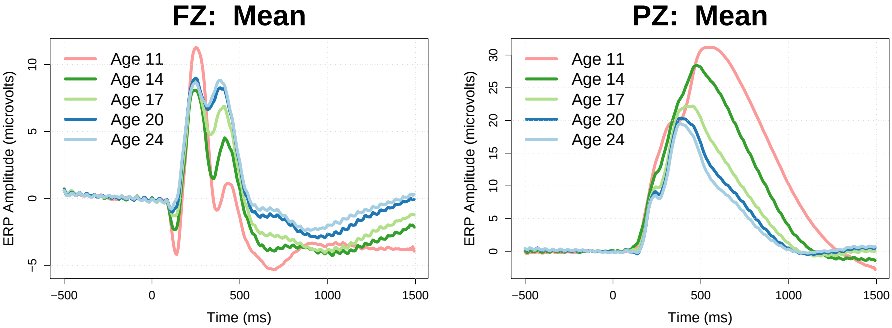

Figure 3: Estimates of the mean ERP $\hat{\mu}$ from fitting the classic ACE model to each combination of time and age.

ACE Estimated Additive Genetic Variance: $\hat{\sigma}_A^2$

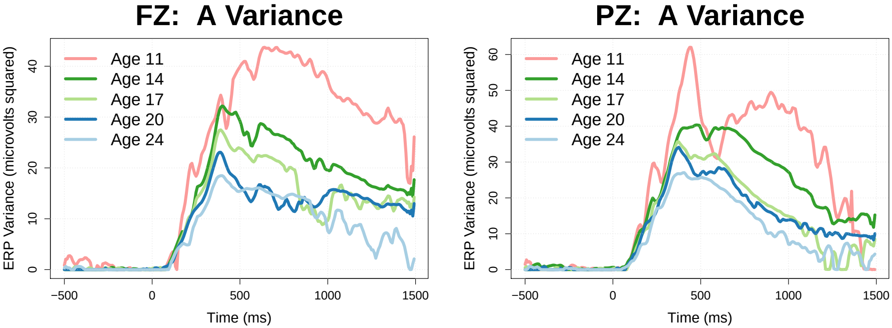

Figure 4: Estimates of the additive genetic variance $\hat{\sigma}_A^2$ from fitting the classic ACE model to each combination of time and age.

ACE Estimated Common Environment Variance: $\hat{\sigma}_C^2$

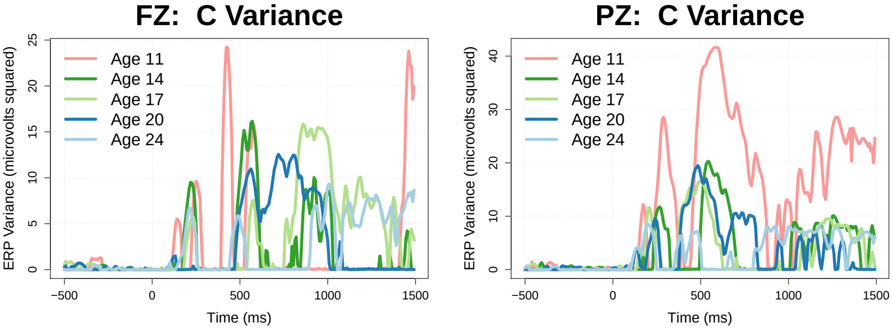

Figure 5: Estimates of the common environment variance $\hat{\sigma}_C^2$ from fitting the classic ACE model to each combination of time and age.

ACE Estimated Unique Environment Variance: $\hat{\sigma}_E^2$

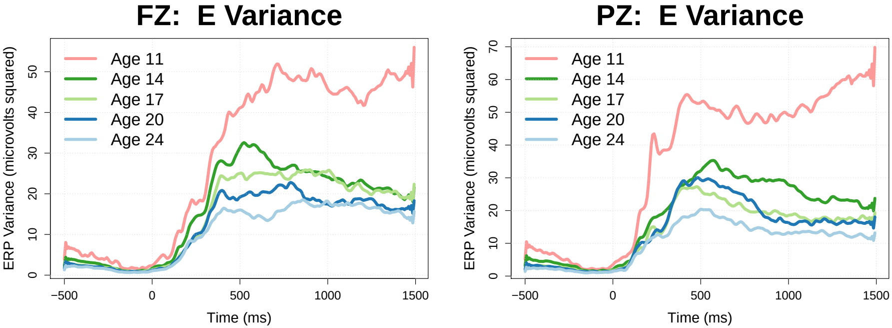

Figure 6: Estimates of the unique environment variance $\hat{\sigma}_E^2$ from fitting the classic ACE model to each combination of time and age.

ACE Estimated Heritability: $\hat{\sigma}_A^2/(\hat{\sigma}_A^2 + \hat{\sigma}_C^2 + \hat{\sigma}_E^2)$

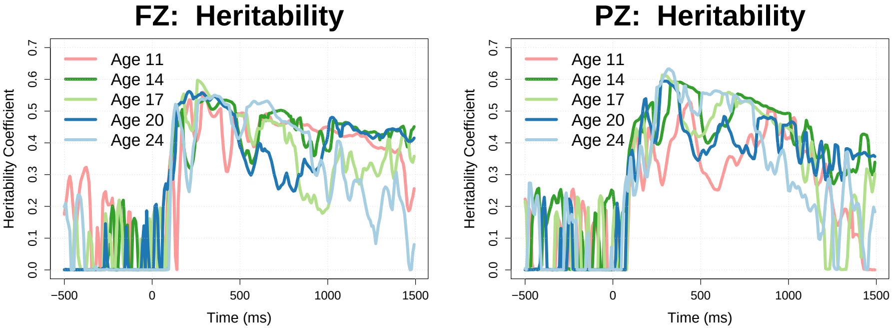

Figure 7: Estimates of the heritability coefficient $h^2 = \frac{\hat{\sigma}_A^2}{\hat{\sigma}_A^2+\hat{\sigma}_C^2+\hat{\sigma}_E^2}$ from fitting the classic ACE model to each combination of time and age.

Functional ACE Model

Functional ACE Model for Twin Data

Let $y_{ijk} \in \mathbb{R}$ denote the k-th measurement of some response variable recorded from the j-th twin of the i-th twin pair.

A functional extension of the ACE model for twin data has the form

$$y_{ijk} = \mu(\mathbf{x}_{ijk}) + a_{ij}(\mathbf{z}_{ijk}) + c_i(\mathbf{z}_{ijk}) + e_{ij}(\mathbf{z}_{ijk}) + \xi_{ijk} \tag{2}$$

where

- $\mathbf{x}_{ijk}$ and $\mathbf{z}_{ijk}$ are known predictor vectors
- $\mu(\cdot)$ is some unknown smooth mean function
- $a_{ij}(\mathbf{z}) \sim \mathrm{N}(0, \sigma_A^2(\mathbf{z}))$ are the latent *additive genetic* effect functions
- $c_i(\mathbf{z}) \sim \mathrm{N}(0, \sigma_C^2(\mathbf{z}))$ are the latent *common environment* effect functions
- $e_{ij}(\mathbf{z}) \sim \mathrm{N}(0, \sigma_E^2(\mathbf{z}))$ are the latent *unique environment* effect functions
- $\xi_{ijk} \overset{\text{iid}}{\sim} \mathrm{N}(0, \sigma^2)$ is white noise (part of unique environment variance)
- $a_{ij}(\cdot)$, $c_i(\cdot)$, $e_{ij}(\cdot)$, and ξ_{ijk} are mutually independent of one another

Covariance Structure of Additive Genetic Effects

The assumed covariance structure for the additive genetic component is

$$\mathrm{Cov}(a_{i1}(\mathbf{z}), a_{i2}(\mathbf{z})) = \begin{cases} \sigma_A^2(\mathbf{z}) & \text{if } z_i = 1 \text{ (monozygotic)} \\ \frac{1}{2}\sigma_A^2(\mathbf{z}) & \text{if } z_i = 2 \text{ (dizygotic)} \end{cases}$$

where $z_i \in \{1, 2\}$ denotes the number of zygotes for the i-th twin pair.

Parameterize the effects as $\mathbf{a}_i(\mathbf{z}) = \mathbf{\Phi}_{z_i}\tilde{\mathbf{a}}_i(\mathbf{z})$ where $\tilde{\mathbf{a}}_i(\mathbf{z}) \overset{\text{iid}}{\sim} \mathrm{N}(\mathbf{0}, \sigma_A^2(\mathbf{z})\mathbf{I})$ and

$$\mathbf{\Phi}_1 = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \qquad \text{and} \qquad \mathbf{\Phi}_2 = \begin{pmatrix} 1 & 0 \\ 1/2 & \sqrt{3/4} \end{pmatrix}$$

are the corresponding matrices for MZ and DZ twins.

Representing the Unknown Mean Function

The predictor vector is $\mathbf{x} = \mathbf{z} = (\text{time}, \text{age})$ where

- $\text{time} \in \{1, \ldots, 256\}$ is the ERP time point (128 Hz data for 2 sec)
- $\text{age} \in \{11, 14, 17, 20, 24\}$ is participant's age

Tensor product smoothing splines (Wahba, 1990) to represent the functions:

$$\mu(\text{time}, \text{age}) = \mu_0 + \mu_T(\text{time}) + \mu_A(\text{age}) + \mu_{TA}(\text{time}, \text{age})$$

where μ_0 is an intercept, $\mu_T(\cdot)$ and $\mu_A(\cdot)$ are the main effect functions, and $\mu_{TA}(\cdot)$ is the interaction effect function.

Cubic smoothing spline reproducing kernels for time and age marginal effects.

Representing the Unknown Mean Function (continued)

For practical computation, note that we can approximate $\mu(\cdot)$ as

$$\begin{aligned}\mu(\mathbf{x}) &= \sum_{v=1}^{m} d_v \phi_v(\mathbf{x}) + \sum_{u=1}^{q} c_u \rho(\mathbf{x}, \mathbf{x}_u^*) \\ &= \boldsymbol{\rho}_M' \boldsymbol{\beta}\end{aligned}$$

where

- $\{\phi_v\}_{v=1}^{m}$ are known functions that span the null space
- ρ is the known reproducing kernel (RK) of the contrast space
- $\{\mathbf{x}_u^*\}_{u=1}^{q} \subset \{\mathbf{x}_i\}_{i=1}^{n}$ are known spline knots
- $\mathbf{d} = (d_1, \ldots, d_m)'$ and $\mathbf{c} = (c_1, \ldots, c_q)'$ are unknown coefficients
- $\boldsymbol{\beta} = (\mathbf{d}', \mathbf{c}')'$ is the combined coefficient vector

Representing the Unknown Random Functions

We could use a similar approach for the unknown random functions, e.g.,

$$\tilde{a}_{ij}(\mathbf{z}) = \boldsymbol{\rho}_A' \boldsymbol{\alpha}_{ij}$$

where

- $\boldsymbol{\rho}_A$ contains known basis functions (e.g., RK functions)
- $\boldsymbol{\alpha}_{ij} \overset{\text{iid}}{\sim} \mathrm{N}(\mathbf{0}, \boldsymbol{\Psi}_A)$ where $\boldsymbol{\Psi}_A$ is the coefficient covariance matrix

However, this approach has some drawbacks…

- $\boldsymbol{\Psi}_A$ could contain many covariance parameters (e.g., 100s or 1000s)
- Need good starting values for an iterative algorithm

Finding an Optimal Basis for the Random Functions

If we knew the covariance matrix (across time and age) for the univariate ACE random effects, we could construct an optimal set of basis functions.

- $\boldsymbol{\rho}_A$ contain eigenfunctions and $\boldsymbol{\Psi}_A$ is diagonal matrix of eigenvalues

Let $\mathbf{S}_A$ denote the BLUP covariance matrix from the univariate ACE model.

- $\mathbf{S}_A$ is 1280×1280 in our case ($1280 = 256$ times points $\times$ 5 ages)
- Smoothed eigenvalue decomposition of $\mathbf{S}_A$ to estimate an optimal basis
- Then use Fisher Scoring algorithm to estimate variance parameters
- Related to the method described in Helwig (2016)

Use a similar approach for the other (environment) random effect functions.

Model Fitting and Tuning

Fit the model in a two stage fashion

- Estimate the mean function $\mu(\cdot)$
- Estimate covariance functions from $\mathbf{r}_i = \mathbf{y}_i - \hat{\mu}(\mathbf{x}_i)$

Two smoothing parameters that need to be tuned

- λ_m controls the smoothness of the mean function
- λ_v controls the smoothness of the covariance functions

Use a version of the Generalized Cross-Validation (GCV) criterion to select the smoothing parameters (Craven & Wahba, 1979; Gu & Ma, 2005).

Generalized Cross-Validation Results

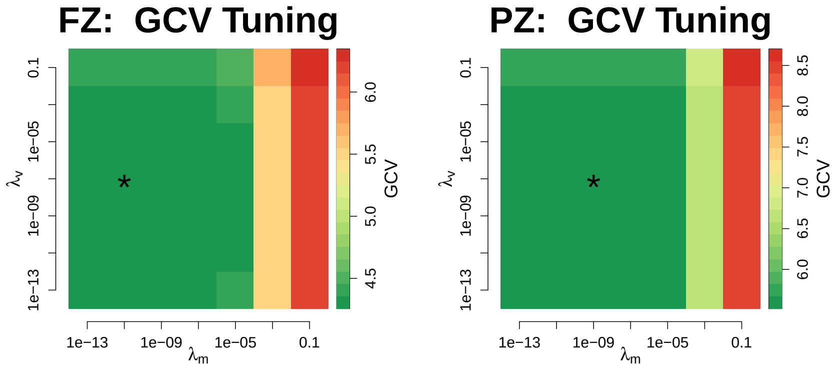

Figure 8: The generalized cross-validation (GCV) criterion as a function of λ_m (mean tuning parameter) and λ_v (variance tuning parameter).

fACE Estimated Mean Structure: $\hat{\mu}(\cdot)$

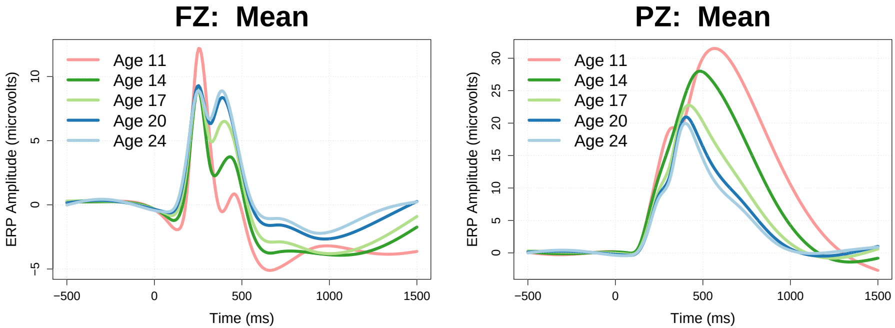

Figure 9: Estimates of the mean ERP $\hat{\mu}(\cdot)$ from fitting the functional ACE model.

fACE Estimated Additive Genetic Variance: $\hat{\sigma}_A^2(\cdot)$

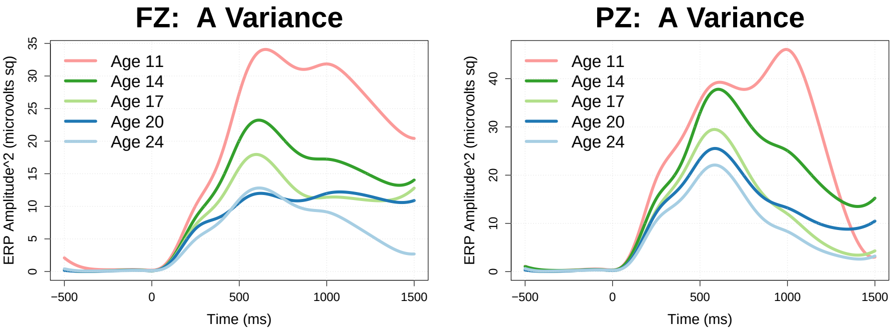

Figure 10: Estimates of the additive genetic variance function $\hat{\sigma}_A^2(\cdot)$ from fitting the functional ACE model.

fACE Estimated Unique Environment Variance: $\hat{\sigma}_E^2(\cdot)$

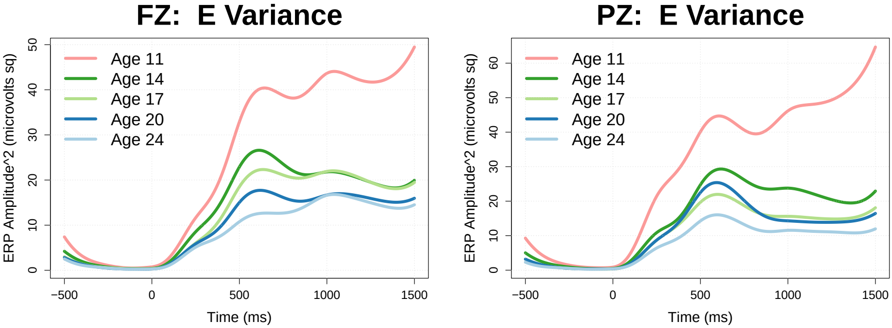

Figure 11: Estimates of the unique environment variance function $\hat{\sigma}_E^2(\cdot)$ from fitting the functional ACE model.

fACE Estimated Heritability: $\hat{\sigma}_A^2(\cdot)/(\hat{\sigma}_A^2(\cdot) + \hat{\sigma}_E^2(\cdot) + \hat{\sigma}^2)$

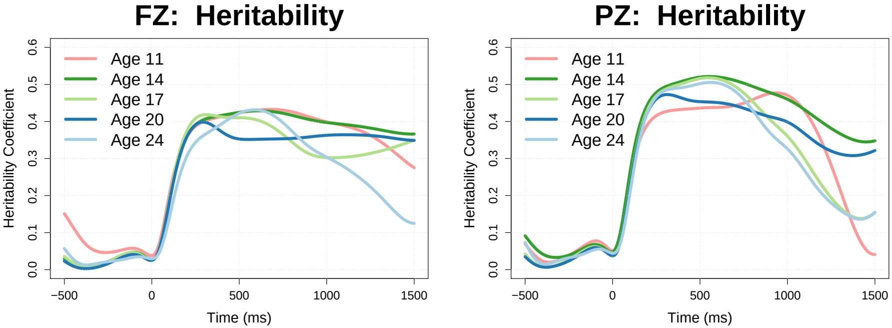

Figure 12: Estimates of the heritability function $h^2(\cdot) = \frac{\hat{\sigma}_A^2(\cdot)}{\hat{\sigma}_A^2(\cdot)+\hat{\sigma}_E^2(\cdot)+\hat{\sigma}^2}$ from fitting the functional ACE model.

fACE Estimated Individual Differences

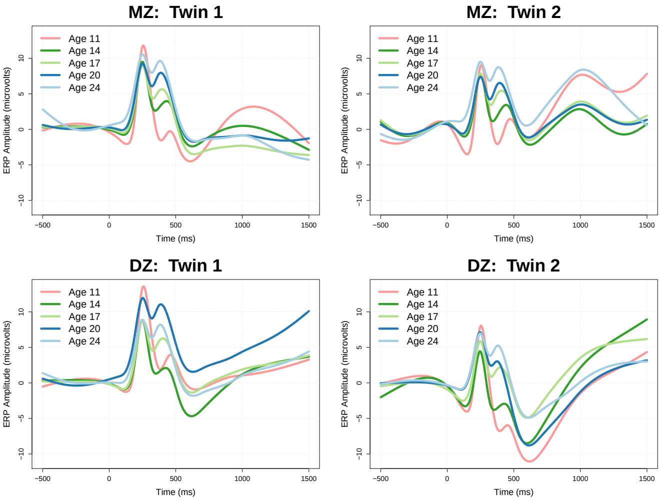

Figure 13: Estimated ERPs for one MZ pair (top) and one DZ pair (bottom).

Conclusions and Extensions

Summary of Findings

Amplitude and form of ERPs change between ages 11–24

- Becomes more compact across development
- Shows more theta activity (3–7 Hz) as we age
- Seems to stabilize by about age 20

Heritability of ERP is much broader than expected

- ERP amplitude is heritable for 1+ sec after stimulus
- What does this mean/imply??

Future Directions of Research

Incorporate more predictors for mean and/or covariance functions

- Task difficulty, response accuracy, sex, etc.
- Substance use/abuse measures—related to P3 amplitude (Iacono & Malone, 2011; Euser et al., 2012)

Constrain $\sigma_A^2(\cdot)$ and $\sigma_E^2(\cdot)$ to change coherently across age

- Heritability should be the same across development
- Introduces functional dependence between $\sigma_A^2(\cdot)$ and $\sigma_E^2(\cdot)$

Smart starting values that do not depend on univariate ACE results

References

Craven, P. and G. Wahba (1979). Smoothing noisy data with spline functions: Estimating the correct degree of smoothing by the method of generalized cross-validation. *Numerische Mathematik 31*, 377–403.

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